Gayathri Rajesh

+1 (408) 663-9097 | g2rajesh@ucsd.edu | linkedin.com/in/rajesh-gayathri | github.com/rajeshgayathri2003

Education

University of California San Diego Sep 2025 – Jun 2027
Master of Science in Computer Science
Relevant Coursework: Probabilistic Reasoning and Learning, Computer Vision I, Introduction to Robotics

National Institute of Technology, Tiruchirappalli Aug 2021 – May 2025
Bachelor of Technology in Electrical and Electronics Engineering ***CGPA: 9.36/10***
Relevant Coursework: Pattern Recognition, Data Structures and Algorithms, Introduction to Data Analysis
Activities and Societies: Robotics and Machine Intelligence, SPIC MACAY NIT Tiruchirappalli Chapter

Experience

National Institute of Technology Tiruchirappalli | *Project Report* Jan 2025 - May 2025
Final Year Project - Guide: Prof. Ankur Singh Rana *Tiruchirappalli, India*

- Implemented a **Game Theory** based approach to determine the optimal selling price for Electric Vehicles; ensuring user satisfaction while maximising charging station profits
- Used reinforcement learning techniques like **Q-Learning**, **Deep Q Networks (DQN)** and **Learning to Search (LEARCH)** to perform EV routing with constraints such as charging time, waiting time and price of charging
- Tested the joint framework for EV routing and pricing on the Sioux Falls Road Transportation Network and achieved a maximum success rate of **88.58%** with DQN.

University of Southern California | *Project Report* May 2024 - Mar 2025
Summer Research Student - Guide: Prof. Daniel Seita *Los Angeles, USA*

- Employed pre-trained foundation models for fabric manipulation tasks like folding and smoothing via robotic arms
- Constructed prompts for GPT-4o, GPT-4 and GPT-4 Vision to generate low-level instructions for manipulation. Examined methods such as fine-tuning and Set-of-Mark prompting to help in visual grounding
- Implemented hand-eye calibration between the camera and bimanual UR5 arms using **MoveIt** and **ROS Noetic**
- Boosted performance by 50% in folding tasks and extended smoothing tasks to non-square fabric by simulating experiments in **SoftGym**

Indian Institute of Technology, Madras | *Project Report* May 2023 - Jul 2023
Summer Research Student - Guide: Prof. Kaushik Mitra *Chennai, India*

- Enhanced extreme low-light videos with illuminance in the 0.1 to 5 lux range. Studied retinex-based, machine learning and deep learning methods including LLNet and U-Net
- Trained the **U-net architecture** on extreme low-light JPG images and achieved an average Peak Signal-to-Noise Ratio (PSNR) of 22.4064 during training and 19.3361 on the test dataset
- Achieved an average PSNR of 21.4331 during training and 14.5273 during testing on the modified U-net architecture designed for faster computation and evaluated the above on the **Oxford Robotcar** Dataset

Projects

Unmanned Underwater Vehicle for Inspection and Repair | *Report* Sep 2023 - Feb 2024

- Built a 5-DOF underwater robot with open frame architecture and identified cracks and corrosion in underwater structures using cameras and computer vision techniques. Leveraged data from IMU and pressure sensors communicated the same to the on-ground station
- Performed transfer learning on VGG16 in **PyTorch** for corrosion detection and achieved an accuracy of 96.55% on the validation set. Used a Convolutional Neural Network for crack detection and deployed them on a **Jetson Nano**
- Finalist in the Smart India Hackathon 2023 organised by the Govt. of India among *2000+* participating teams

Sight Stick | *Report* Jan 2023 - Apr 2023

- Designed an intelligent blind stick that can assist the visually impaired by detecting objects and providing haptic feedback using a vibrating wristband
- Implemented an object detection algorithm using **TensorFlow Lite** and **OpenCV** on a **Raspberry Pi 4**
- Devised features for additional safety and emergency assistance using a mobile application connected to the stick. Interfaced the Pi to the wristband using an HC05 Bluetooth module
- Finished among the *top 5* from *100+* teams at Sangam, a national-level hardware hackathon organised by NIT Trichy

Technical Skills

Programming Languages: Python | C/C++ | Matlab | GNU Octave | Arduino C
Libraries and Tools: PyTorch | OpenCV | NumPy | Pandas | Matplotlib | Keras | ROS2 | SQL | Linux | Git
Programmable Boards: Arduino | Raspberry Pi | Jetson Nano | ESP 32

Honours and Awards

- Received the **Micron URAM Scholarship 2023-24**, based on brilliance in STEM fields and willingness to contribute to society. Chosen in the top 60 students among thousands of applicants and awarded a cash prize for the same
- Won the **Jitheshraj Scholarship for Promising Freshmen** (2021-22 cycle), among 1200+ candidates, based on an initiative to attain technological awareness and generate positive change in society